

**Fox Brothers (Leyland) Ltd**  
**SAFETY PROCEDURES**  
**EXCAVATIONS AND TIPS RULES**  
**TONG QUARRY, BACUP**  
**July 2025**

**Reference should be made to available drawings and the Geotechnical Assessment**

## **1. PURPOSE**

To establish the Fox Brothers (Leyland) Ltd for suitable and sufficient rules for the safe construction and operation of excavations and tips at Tong Quarry.

## **2. INTRODUCTION**

These rules are essential for the management of excavations, tips (including product stockpiles), in pit backfill and inert imported material waste tip are practical measures for keeping excavations, tips and people on and around them, safe. The rules cover excavation work, tip and product stock construction and removal, including lagoons and any in-pit backfill and should be read in conjunction with the geotechnical assessments.

## **3. LEGAL FRAMEWORK**

Employers must provide a safe place of work to meet the general requirements of the Health and Safety at Work etc Act 1974. The Quarries Regulations 1999, Regulation 31 "Excavations and Tips Rules" has a specific requirement for the preparation of these rules as does the "Work at Heights Regulations 2005".

## **4. IMPLEMENTATION**

The following sections outline the procedures to be followed in relation to Tips, Excavations, Lagoons and Stockpiles:

Section 5 – Provision and Maintenance of Excavations

Section 6 – Construction, Maintenance and Removal of Tips (including stockpiles)

Section 7 – Stockpiling Grounds

Section 8 – Site Inspections

Section 9 – Plant and Machinery

Section 10- Supervision

Section 11 -Review

The above sections should be read in conjunction with the Quarry Manager's Rules and the geotechnical assessments.

## **5. PROVISION & MAINTENANCE OF EXCAVATIONS**

### **5.1 Excavation Designs – Quarry**

#### **THE MAXIMUM DEPTH OF EXCAVATION – Approximately 20 metres**

##### **Excavation Profile**

The excavation profile should be in accordance with the approved design and comprise the following:

- I. A standoff to the overburden slopes should be 2m to allow for inspection of the crest and reduce rockfall / material being washed over the face.
- II. The overburden is to be left at 34°.

- III. The first 2m to 3m of weathered rock can be at 34° but the next 6m can be at 75/80°.
- IV. Then there should be an intermediate horizontal bench of a minimum of 3 metres in width at 8 to 10 metres below ground level. This should have a rocktrap constructed 1m high and 1m from the open edge.
- V. The lower face 8 to 10 metres to the base of the Lower Mountain Mine can be a single excavation face inclined at 75/80°.
- VI. Another 3m wide bench with a rock trap is then created.
- VII. The Bottom face 6 to 8 metres to top of mudstone horizon - single excavation face inclined at a gradient of 1 vertical to 0.5 horizontal, (i.e. 62.5° from horizontal).
- VIII. Intermediate mid-way internal working operating benches must be at least 15 metres in width making provision for edge protection on the outer edges and rock gutters adjacent to faces, in compliance with the requirements of Regulation 13, i.e. minimum height 1.5 metres for edge protection. See appended sheet for guidance on rock gutter design requirements.

The edge protection bund should be constructed in compliance with the recommendations detailed in Regulation 13 (appended to these Rules).

Adjacent to excavated faces a suitable rock gutter / Rock Trap should be constructed.

If it is proposed to vary the above profile it should only be done following an approved (by the Geotechnical Specialist) revised design being specified.

## **5.2 Excavation Design – Stockpiles**

Excavation from stockpiles must be undertaken from faces with a height which do not exceed the maximum reach of the plant being used. Where stockpiles are constructed to a height which exceeds this limit, working should be undertaken by either

- a) Periodically dozing down the feed face height such that it complies, or
- b) Work the stockpile in two lofts, thereby splitting the height.

## **5.3 Excavation Rules**

The following measures should be adopted as being necessary during the excavation and maintenance of excavation faces:

- (i) Excavation in soils and overburden should not exceed the consented limit of excavation at surface without the express approval of the Quarry Manager.
- (ii) Excavations should have regard to the batters and benches required to maintain the stability of the overlying strata in order to ensure that the overall slope, in general, does not fall outside that agreed by the Geotechnical Specialist and the Quarry Manager.
- (iii) Excavation slopes in superficial deposits should be designed to ensure stability. The design for this will be agreed between the Geotechnical Specialist and the Quarry Manager.
- (iv) Safety benches in rock should be a minimum width of 3 metres and located at rockhead (or base of weathering) and at vertical intervals or as otherwise defined by the Geotechnical Specialist and agreed with the Quarry Manager.
- (v) If at any point, changes in design could fall outside the parameters laid down in the geotechnical design then the Geotechnical Specialist must carry out a further appraisal/ assessment to ensure that the new design will still afford a safe environment. This must be implemented before any works are undertaken on site.

## **5.4 Excavation Operator Rules**

- (i) Excavator operators should ensure that overhangs are not left and that faces are dressed to remove loose material as they are developed. No face dressing from the overlying bench is permitted.
- (ii) Where a 360° excavator is used and unless authorised by the Quarry Manager or the General Foreman, the height of the working face should never exceed the reach of the bucket of the excavation plant being utilised.
- (iii) The excavator operator should ensure that a safe working area is prepared commencement of operations as directed by the Supervisor.
- (iv) The excavator / dozer operator should ensure that the loading area is maintained free of spillage at all times.
- (v) All loading should be undertaken on the excavator driver's unobstructed side.
- (vi) Whilst loading is underway, no vehicle should be parked within the swing radius of the excavator bucket or reversing circle of the front-end loader other than for purpose of being loaded.
- (vii) If used, when dumptrucks (ADTs) are being loaded, parking brakes shall be applied, and the vehicle driver must remain in his cab.
- (viii) Any excavation work in the vicinity of overhead lines shall be in accordance with HSE GS6
- (ix) The excavator operator should ensure that at the end of the shift the excavator is moved away from the excavation face and parked on level ground.
- (x) No pedestrian should enter the rock trap from the toe or the crest of the wall unless authorised to do so by the Quarry Manager. Authorisation shall be limited to essential duties only, e.g. surveying, supervising, pump attendance etc, and such persons shall be authorised after a suitable toolbox talk.
- (xi) All machines working / driving in the 5-metre zone from the toe of the wall should have FOPS protection on the cab.
- (xii) Any areas cordoned off for instability reasons should not be entered without a permit to work.
- (xiii) No person or machine should enter any safety bench without a permit to work.
- (xiv) Only authorised persons should stand on (the inside edge of) edge protection bunds, for the purpose of site inspections only. Fenced vantage points should be provided for other viewing purposes.
- (xv) Vehicular and pedestrian access to the crest of the excavation and to safety benches should be prohibited with suitable barriers.
- (xvi) Any unsafe development during a shift, operators to stop work move to a safe area and liaise with Quarry Manager.

## **5.5 Excavation Drainage Rules**

- (i) Cut-off drainage should be provided as necessary around the excavation area to control surface water run-off into the excavation area.
- (ii) Water should not be allowed to accumulate, as far as is reasonably practicable, in the excavations except in sumps as agreed with the Quarry Manager.
- (iii) Where required, pumps should be provided at sumps to prevent water accumulating in the excavation.
- (iv) Pumps should be located, where practicable, at least 10 metres away, from the toe of the wall.
- (v) Where required, pumping lines should be provided from the excavation sump to the water treatment system to prevent water accumulating in the excavation.

- (vi) Intercepting drainage ditches should be provided as necessary in the excavation area to direct water to the temporary sumps and away from the working places.
- (vii) Pumping should be undertaken in such a manner so as to ensure that drainage ditches are not damaged, and that adjacent land and property is not flooded.
- (viii) Measures should be provided to ensure that pumped water is not allowed, as far as is reasonably practicable, to re-enter the excavation area.
- (ix) Pumped water should not be pumped into areas of backfill unless a specific geotechnical assessment has been undertaken beforehand.

## **5.6 Excavation Haul Road Rules**

- (i) Suitable haul roads and ramps from the excavation area to the primary stockpiling or tipping location should be provided to ensure the safe passage of dumptrucks / FELs. Adequate turning areas should be provided at the points of excavation and tipping.
- (ii) Haul roads, where necessary, should be provided with suitable drainage side channels, ditches, culverts and slurry pits and edge protection.
- (iii) Where two-way traffic is permitted, main haul roads should have a minimum width of 3.5 times the width for 2-way traffic and a maximum gradient of a max of 1v:8h for wheeled vehicles and 1v:5h for tracked or as agreed by the Quarry Manager. Waste tip access should be 1 in 10 (approx.)
- (iv) Other roads should have a minimum width equivalent to that of the largest vehicle width authorised to use the road plus 2 metres (i.e. approximately 5 metres) and a suitable gradient as determined by the Quarry Manager.
- (v) Haul roads and turning areas should be maintained to minimise standing water and slurry at all times.
- (vi) Haul roads and turning areas should be maintained with suitable cambers or crossfalls by re-surfacing as necessary and by the regular use of front-end loader / dozer to remove soft spots and ruts.
- (vii) All haul road edges should be protected by a bund of material 1.5 metres high or half the diameter/height of the largest truck tyre utilising the road, whichever is the greatest, at all times. The bund should be substantially constructed i.e. a minimum of 1.5 metres wide (on top of the bund) or as directed by The Quarry Manager to ensure that FELs or waste delivery vehicles cannot pass over the edge of the excavation/tip.
- (viii) All haul roads should be watered to control dust emissions. Spot spraying on ramps, bends etc. should be undertaken to prevent vehicles from losing traction.
- (ix) Additional edge protection should be considered in high-risk areas such as sharp bends and the bottom of a ramp.
- (x) Haul roads constructed adjacent to rock faces should be positioned at sufficient distance from the face to enable the construction of a rock gutter. An alternative to a gutter below operating levels would be the construction of an earth bund on the outer edge of the rock catch area.

## **5.7 Supervision**

The excavation operations shall be supervised by the Site supervisor on a daily basis with the Quarry Manager having direct control over the operations.

Any problems encountered during excavation operations shall be immediately brought to the attention of the Quarry Manager.

## **5.8 Excavation Inspection Rules (Regulation 12)**

Daily working place safety inspections shall be completed by a competent person identified in the management structure for the site and findings shall be recorded on the "quarry inspection report" pro-forma. Any defects shall

be brought to the attention of the Quarry Manager forthwith.

A pre-shift commencement inspection of all working areas and access routes must be undertaken by an appointed person and recorded. Any defects must be reported immediately to the Quarry Manager, affected area(s) isolated and problem rectified before operations re-commence in the area. Where necessary the Geotechnical Specialist should be consulted with respect to the nature of remedial works and compilation of a suitable **Permit to Work** (Regulation 18)

Inspection reports shall be kept in accordance with the Health & Safety Document and kept on site or at head office.

### 5.9 Suspension of Excavation Operations

Excavation operations shall be suspended during adverse weather (amber met office warnings) weather at the discretion of the supervisor / person making inspection before excavation operations recommence, an inspection of the work area shall be completed to assess that the tip is in a secure condition before restating work.

### 5.10 Amendment to Excavation Rules

Only the Quarry Manager, in consultation with the Geotechnical Specialist, shall amend the Excavation Rules. In the event of any changes, e.g. in material being excavated / method of excavation / supervision / inspection, then the Quarry Manager shall be informed as soon as practicable prior to implementations of that the rules can be amended, approved by the Geotechnical Specialist and re-issued to the persons concerned.

## 6. CONSTRUCTION, MAINTENANCE AND REMOVAL OF TIPS (INCLUDING STOCKPILES)

The following measures should be adopted as necessary before, during and after the construction of any tips, including lagoons. Maintenance is of paramount importance and guidance is given throughout the following paragraphs:

### 6.1 Materials Classification and Impact upon Tip Design and Formation Method

The table below provides a broad guidance with respect to the types of materials which will be excavated in UK quarry, opencast coal, brick clay and most other industrial minerals extraction sites. The classification is intended to provide guidance for site personnel involved in the formation of tip structures.

| CLASS | MATERIAL TYPES                                                                                       | METHOD OF DEPOSITION                                                                                                |
|-------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1     | Competent un-blasted mudstones, siltstones and occasional sandstones                                 | Edge tipping or block tipping                                                                                       |
| 2     | Stiff boulder clay<br>Weathered bedrock                                                              | Edge tipping, tipping short of edge and block tipping                                                               |
| 3     | Soft imported inert waste<br>Dust from stone crushing operation                                      | Tipping short of edge or block tipping                                                                              |
| 4     | Slurry dredged from settling lagoons, drainage ditches and sumps.<br>Material graded off haul roads. | Method to be determined following risk assessment. This could include the creation of bunded areas with the tip (s) |

Materials contained within the Tong Quarry, and which are likely to be tipped, would broadly fall into Classes 2, 3 and 4, with the most significant proportion being Class 3, i.e. imported waste, which is currently tipped in the western (exhausted) quarry void.

### 6.2 Imported Inert Waste Tips Design

The tip design requires that material is layered and compacted, this construction design is inherently more stable than end tipped designs.

- In order to ensure stability, and to reduce the effects of bulking, the tipped materials shall be block tipped in layers of 1 metre, which are to be individually compacted by at least 4 passes of the emplacing bulldozer.

- Rock traps must be constructed to maintain a standoff of 5m from any quarry slopes to a minimum height of 1.5m.
- Any standing water should be drained or pumped from the area of construction.
- Any superficial soils and soft clays/mudstones should be removed from beneath the footprint of the ramp; any significant depressions should be filled with free draining rock.
- The drainage blanket under the tip should be no less than 1m thick and formed using coarse, clean rockfill material.
- The final (and in situ for more than a week) outer slopes of the tip should not exceed 1v:4h (14°).
- Temporary slopes in situ for up to a week and less than 10m high can be dozed to 1v:2.5h (22°).
- The slope angles should be monitored and controlled by a competent person (the designated supervisor) using batter boards, an inclinometer level or other similar device such that the designed profiles are achieved.
- Material should be tipped on stable, level or gently sloping ground such as to not cause a danger of overtipping when the dump truck skip or HGV body is raised.
- Suitable flat turning areas of sufficient size must be created to ensure that dump trucks (and other plant) can safely manoeuvre.
- When tipping onto the top of a tip, the dump truck or HGV may only proceed if full edge protection is in place. Edge protection shall be maintained where all wheeled vehicles operate and bunds should be formed in accordance with the requirements of Regulation 13 of the Quarries Regulations 1999. Drainage pipes or small gaps should be left at regular intervals to facilitate surface water runoff.
- Dump trucks should always tip short (minimum 10 metres) of any open edge of the tip. DO NOT end tip material over a tip face or layer edge.
- Wet or weak material should be separately stored so that it can drain and partly desiccate, before it is intermixed with more competent material and deposited within the main body of the tip or in its upper parts (i.e., where it is unlikely to compromise stability).
- Tipping should be suspending during adverse weather conditions or poor visibility. If the tip is saturated then all tipping operations should be suspended. Adverse weather includes snow, high rainfall (more than 25mm in 24 hours), frost etc.
- No frozen material shall be placed or incorporated in the fill.
- Material tipped on the placing surface should be spread and compacted as soon as possible and in any case before the end of the working day.
- Each layer should be completed and fully compacted prior to the next layer being placed.
- Every effort must be made to key or step the layers into each other to avoid forming significant planes of weakness.
- The tip should be accessed using ramps and haul roads formed ideally at a grade of no steeper than 1v in 8h (7°) and with edge protection of no less than 1.5m height.
- All tipping operations should be carried out behind edge protection of no less than 1.5m in height.
- The surface of the tip should be domed or adequately graded, and gaps left in edge protection to prevent any ponding of water on the surface.
- The volume and type of material deposited or removed should be recorded under Regulation 36.
- A minimum standoff of 5m must be maintained between the toe of the tip and any excavations.

- Drainage along the faces can be provided by tipping competent coarse material along the face as the tip is built up.
- Wet material should be placed in cells constructed with 3m high bunds and allowed to consolidate / dry out prior to being overtipped by capping material, and their positions recorded.
- It is a requirement of this design to closely monitor the stability of the tip slopes during and post construction. The tip should be inspected by an authorised competent person in the management structure each day before tipping takes place. The daily inspections should cover the stability and working practices within the area of active tipping, in addition to the stability of all existing tip slopes and rock faces in the vicinity of all tipping activities and access routes.
- If the inspections undertaken by quarry personnel detect any evidence of instability or if there is any cause for concern, then further geotechnical advice should be sought immediately. If such inspections identify an imminent risk of danger within the working area or along any roadways, then all operations and access within the area should be suspended until further geotechnical advice is undertaken and appropriate remedial measures undertaken.

The stability analysis is to be found in Appendix 5 and shows that if the tip is constructed to this specification, it will be stable.

## **6.3 Method of Tip Construction**

### **6.3.1 Tip Foundation Rules**

- (i) Tips must not be constructed over any known spring or watercourse unless suitable under-drainage has been established.
- (ii) Any spring or watercourse encountered in the course of tipping should be reported to the Quarry Manager immediately and diverted or culverted subject to the Manager's approval before tipping is continued in that area.
- (iii) Under-drainage should be provided and maintained where necessary as designed by the Geotechnical Specialist, to ensure the stability of the tip.

### **6.3.2 Tip Construction Rules**

- (i) No tip should be constructed either to a height or in such a manner as to constitute a danger to persons or property on the site or in its vicinity.
- (ii) No tip should be constructed in such a manner as to pose the risk of damage to fences, walls, or hedges, on the perimeter of the site.
- (iii) Backfill / imported waste materials shall not be placed on an adversely dipping surface unless otherwise instructed by the Site Manager. The Site Manager shall in any case seek expert advice from the Geotechnical Specialist before any such operation is carried out.
- (iv) All areas of standing water shall be removed by providing suitable drainage or pumping to a suitably constructed sump. Water shall be regularly removed by pumping from this sump into the site's drainage system.
- (v) Tips should [where practical] be constructed from the quarry floor upwards. End tipping must be avoided, though tipping and dozing down the tip face is acceptable.
- (vi) Tips should be constructed in layers, with material being deposited in layers of not more than 1 metre in thickness, with falls away from the edges to ensure adequate consolidation and drainage control.
- (vii) When placing backfill in the void the maximum free face height in general should not exceed 10 metres unless a specific block tipped in layers of 1 metre" so am not sure what this means. GA says Temporary slopes in situ for up to a week and less than 10m high can be dozed to 1v:2.5h (22°).

(ix) No tipping into water should be permitted.

(x) Tips should be constructed with face slopes at or less than the safe angle of repose for the material / soils being deposited.

(xi) As construction of the tip progresses, where necessary the profile should be graded and trimmed, so as to present a smooth and uniform appearance without ragged tops, edges, and slopes.

(xii) The height of the tip should be restricted by utilising the maximum area designated for tipping and be regularly monitored to ensure that the maximum height is not exceeded, as set out by the Engineering Manager.

(xiii) Prior to tip construction, the excavation pavement or newly formed tipping area shall be inspected to ensure that no weak materials are present which may act as a potential failure plane. If present, these materials shall be removed by excavation. The excavation pavement may have to be ripped, dependent on the ground conditions.

(xiv) All above ground tips shall have safety benches with a safety bund/berm of material 1.5 metres high or half the diameter/height of the largest vehicle tyre, whichever is the greatest, if vehicular access is required.

### **6.3.3 Tip Drainage Rules**

(i) Surface water drainage should be diverted away from the base of any tip by suitable cut-off ditches.

(ii) The tops of tips should have a well compacted surface and rounded-off such that all surface water run-off is collected and rapidly discharged.

(iii) If deemed appropriate by the Geotechnical Specialist principal drainage channels should be armoured with suitable stone or geotextile to the approval of the Quarry Manager.

(iv) Tip haul roads should have a cross-fall to enable the efficient shedding of run-off into the site drainage system.

(v) Drainage measures should be established progressively during the construction of the tip to ensure stability and minimise erosion.

### **6.3.4 Tip Haul Road Rules**

(i) Refer to Clauses 5.6 (i) to (ix).

(ii) Roads should not be located below a tipping face, unless unavoidable, and in such cases should be protected by a suitably constructed rock trap and / or edge protection as determined by the Quarry Manager.

(iii) Rock traps should be placed at an adequate distance from the toe of the tip face, depending on the nature of the tip material and the height of the slope.

(iv) The size and construction of the rock trap shall be determined by the Quarry Manager, depending on the nature of the tip material and the height of the slope.

(v) The edge protection shall be increased in size, if necessary, as determined by the Quarry Manager, if it is to serve as a rock trap.

### **6.3.5 Tip Operator Rules** **Dozer Operations**

(i) Any Class 4 materials shall be confined to the upper part of the tip structure.

(ii) Any wet Class 4 materials shall only be deposited in an area designated by the supervisor in charge of the operation (e.g. where at least partial drainage can be achieved prior to intermixing and spreading with drier or more competent fill materials, or where they can be suitably confined so that they do not give rise to instability) and after the required risk assessment.

- (iii) Every effort shall be made to impart compaction by trafficking areas of backfill and stockpiling.
- (iv) Edge protection to a MINIMUM height 1.5m or the radius of the largest wheel operating on the tip, whichever is greater, must be maintained where there is an adverse slope (e.g. along haul road access and on the advancing tip edge) or some other potential hazard using a dozer or dozers.

If for any reason this cannot be maintained along the tip edge (e.g. due to its failure), then the waste delivery vehicles shall deposit material no closer than 6m from the edge and under the supervision of a dozer driver (who must be in attendance at all times) so as to form a safety bund for future tipping operations.
- (v) To facilitate tipping, the delivery vehicles are allowed to pull-off during discharge but must return the skip/body to its travel position as soon as possible.
- (vi) The tip edge shall be regularly examined during the shift for signs of deterioration by the tip supervisor. If the tip edge becomes unstable, the area shall be blocked off to prevent vehicle access. The Site Manager shall be informed as soon as is practicable so that appropriate remedial measures can be taken.
- (vii) A rock-trap of not less than 3m in width and 1.50m in depth shall be maintained along the toe of the slopes as the level of fill is raised and the tip is extended laterally, or as otherwise instructed by the Site Manager.
- (viii) Measures shall be taken to ensure that materials do not roll into any working areas by providing adequate stand-offs and, if required, temporary bunds of not less than 1.50m in height.
- (ix) If it is found that the materials are of a poorer quality than was expected or adversely affected by the excavation method and / or the weather conditions, then the means by which they are deposited may be changed at the discretion of the Site Manager or the supervisor in charge of the operation.
- (x) Tipping up to the final quarry sidewalls must ensure that an adequately sized rock gutter adjacent to the face is retained at all times. For reference the gutter width should be equivalent to approximately 25% of the face height.

#### **6.3.6 Edge Tipping**

- (i) These rules concern the tipping of Class 1 materials on or over the edge protection bund. They may also include Class 2 materials as directed by the Site Manager or the supervisor in charge of the operation.
- (ii) Edge tipping shall only be carried out where suitable edge protection can be formed and maintained at a height of not less than 1.50m.
- (iii) FELs / waste delivery vehicles shall approach the tip area at a safe distance of at least the width of the tipping vehicle from the edge protection, to enable the drivers to examine the area before reversing
- (iv) All tipping vehicles shall reverse square to the tip edge when tipping and keep the dozer, if present, on the driver's side where practicable.
- (v) No waste delivery vehicle shall reverse further than the edge protection and under no circumstances shall they mount it.
- (vi) The manner of any subsequent tipping operations shall be determined by the Site Manager or the supervisor in charge of the operation.

#### **6.3.7 Tipping Short of the Edge**

- (i) These rules concern Class 2, 3 and 4 materials as directed by the Site Manager or the supervisor in charge of the operation.
- (ii) Delivery vehicles shall approach the tip area at a safe distance of at least the length of the tipping vehicle from the edge protection, to enable the drivers to examine the area before reversing.

- (iii) All tipping vehicles shall reverse square towards the tip edge when tipping and keep the dozer on the driver's side where practicable.
- (iv) Delivery vehicles shall deposit their load not less than 5m from the tip edge under the supervision of a dozer or FEL driver and the material shall then be pushed forward using that machine.
- (v) Dozer / FEL operators must satisfy themselves that it is safe to push material over the tip edge both prior to and during tipping operations, ensuring that the machine operates at right angles to the tip edge.
- (vi) The manner of any subsequent tipping operations shall be determined by the Site Manager or supervisor in charge of the operation.

### **6.3.3 Block Tipping**

- (i) These rules concern Class 2 and 3 materials as directed by the Site Manager or the supervisor in charge of the operation.
- (ii) Delivery vehicles shall deposit their loads as individual piles or strips respectively within an area designated by the Site Manager or the supervisor in charge of the operation.
- (iii) If it is required to construct edge protection, then the delivery vehicles shall deposit their loads no closer than 5m from the edge under the supervision of a dozer driver. A safety bund shall then be formed by the dozer for subsequent tipping operations or alternatively ranging poles installed.
- (iv) Delivery vehicles shall not mount the edge protection, nor shall they tip material over or onto it.
- (v) The upper surface of the block tipped material shall be levelled using a dozer before any additional material is deposited.
- (vi) If it is required to push any material over the tip edge, then the dozer operator must satisfy himself that it is safe to do so, ensuring that the machine operates at right angles to the tip edge.
- (vii) The manner of any subsequent tipping shall be determined by the Site Manager or supervisor in charge of the operation.

### **6.3.9 Tip Inspection Rules**

- (i) All tipping areas should be inspected regularly throughout the day by the appointed person.
- (ii) Recorded inspections of tips under construction should be daily. Recorded inspection of completed tips should be weekly.

## **7. STOCKPILING GROUNDS**

### **7.1 Foundation Rules**

- (i) Stocks must not be constructed over any known spring or watercourse unless suitable under-drainage has been established
- (ii) Any spring or watercourse encountered in the course of tipping should be reported to the Quarry Manager immediately and diverted or culverted subject to the Manager's approval before tipping is continued.
- (iii) Under-drainage should be provided and maintained where necessary as designed by the Geotechnical Specialist, to ensure the stability of the stocks.

## **7.2 Stockpile Construction Rules**

- (i) No stock should be constructed either to a height or in such a manner as to constitute a danger to persons or property on the site or in its vicinity. The maximum height of any stock will not exceed 10 metres unless a specific design has been produced by the Geotechnical Specialist.
- (ii) No stockpile should be constructed in such a manner as to pose the risk of damage to fences, walls or hedges, on the perimeter of the site.
- (iii) Where possible stocks should be formed on level ground with the access for tracked vehicles only, 1v:8h for tyred FELs. Where stocks are formed downhill, loads should be tipped short and material pushed over the tipping edge.
- (iv) The running width of the access ramp onto the stocks should be a minimum of twice the width of the largest vehicle to use it.
- (v) As construction of the stock progresses, the ramp should be graded and trimmed so as to present a smooth and uniform appearance.
- (vi) Suitable edge protection will be provided along the full length of the access ramp and at the tip edge. The bund of material should be a minimum height of half the diameter /height of the largest wheel and a minimum width of 2 metres. Vehicles can only use stockpiles which have 1.5m of edge protection to define a No- Go area for FELs or ADTs or other wheeled plant.
- (vii) Where dumptrucks are used, there is to be no over-edge tipping operations.
- (viii) The height of the tip should be restricted by utilising the maximum area available for tipping and should be regularly monitored to ensure that the maximum height of 10 metres is not exceeded unless specifically designed by the Geotechnical Engineer.
- (ix) The tip should be progressively constructed to maximise consolidation by the trafficking of the FEL and the dozer.
- (x) All stockpiles should be built in accordance with the standard construction methodology as defined below.

## **7.3 Product Stocks Drainage Rules**

- (i) Surface water drainage should be diverted away from the base of any stocks by suitable cut-off ditches.
- (ii) Stockpile haul roads should have a cross-fall to enable the efficient shedding of run-off into the site drainage system.
- (iii) Drainage measures should be established and maintained progressively during the construction and life of the stockpile.

## **7.4 Stockpile Operator Rules**

- (i) Tipping directly over the tip edge where there is no bund is strictly prohibited.
- (ii) All FELs should approach at right angles towards to the edge as should front-end loaders.
- (iii) No FEL should advance further than the tip edge protection.
- (iv) The dozer or FEL operator working in the stocking ground should be responsible for observing the safety of the stock edge at all times.
- (v) If the stocks become unstable the FEL driver should immediately block off the tipping area and report the incident to the Supervisor.

(vi) During the hours of darkness, sufficient illumination should be available to enable tipping operations to be carried out in a safe manner. Whilst maintaining the tip the wheel loader operator must regularly check that the edge protection is suitable and sufficient.

(vii) All FELs are to approach the edge protection at the end of the ramp at an angle of 80 to 90 degrees. Under no circumstances are the rear or front wheels of the FEL to mount the edge protection.

## **7.5 Stockpile Removal Rules**

(i) Material is not to be removed from the foot of the stocks whilst an FEL is operating on top of the stocks discharging its load if such operations cannot be segregated.

(ii) The stocks should be removed in such a way that the natural angle of repose at the face of the stock is not dramatically affected. Where necessary, the loader operator should reduce the total height of the stocks to ensure a stable face.

(iii) Where necessary (i.e. where height exceeds the reach of the loading-out plant), the stockpile should be split into two working levels.

## **8. SITE INSPECTIONS**

(i) All inspections should be in accordance with the inspection scheme for Excavation Faces, Spoil Heaps, Lagoons and Watercourses.

(ii) Regular meetings should be held between the Quarry Manager and the General Foreman (or his deputy) to ensure that construction is proceeding in line with the approved design. A record must be maintained of any matters arising that may impact upon the design and / or necessitate a change from the original design. This record must include any necessary instructions to be passed to the relevant operators

(iii) The Geotechnical Specialist will undertake 24 monthly assessments at the site and will work with the Quarry Manager to ensure that work has progressed across the site in line with the original designs. In the event of a change in design, then these must be agreed by the Quarry Manager and Geotechnical Specialist and a record shall be retained (signed by each party) to show any alterations from the original design.

## **9. PLANT & MACHINERY**

(i) Hydraulic excavators and wheel loaders of suitable size and construction shall be used to excavate drift material, overburden and rock.

(ii) Wheel loaders shall be used for loading product from stockpiles with excavators being used to load into the crusher.

(iii) Front-end loaders of suitable size and construction shall be used to transport the above excavated material to the stockpile areas and from stockpiles to the processing plant.

(iv) A dozer or front-end loader of suitable size and construction shall be used to construct and maintain loading / tipping areas and roads on the inert waste tip.

## **10. SUPERVISION**

(i) The Quarry Manager should appoint a competent person(s) responsible for supervising the excavation and tipping operations for each operational shift.

(ii) The appointed Supervisor should undertake daily inspections of the excavation faces, tipping faces, edge protection, rock traps and haul roads.

(iii) The Supervisor should ensure that vehicle logbooks are completed daily to monitor the plant operator's attention to vehicle safety.

## 11. REVIEW

These Excavations and Tips Rules should be reviewed on an annual basis with reviews being counter-signed by the Quarry Manager and Geotechnical Specialist.

Date of next review July 2026

Quarry Manager Name .....Barry Thompson.....

Quarry Manager Signature .....*B Thompson*.....

Date .....24/07/2025.....

Geotechnical Specialist Name Ian Thomas

Geotechnical Specialist Signature



Date .....25/07/2025.....